

Econ 101A

Section 6

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1 Preferences and Utility Functions

1.1 Properties of Preferences

- $\succsim$ is **complete** if for all x and y in X either $x \succsim y$, or $y \succsim x$ or both.
- $\succsim$ is **transitive** if for all x, y , and z , $x \succsim y$ and $y \succsim z$ implies $x \succsim z$.
- $\succsim$ is **rational** if $\succsim$ is *complete* and *transitive*.
- A rational preference $\succsim$ is **continuous** if for all $x, y \in X$ such that $x \succ y$, we can find some $\epsilon > 0$ such that for all x' and y' that are less than ϵ distant from x and y , respectively, we have $x' \succ y'$.
- $\succsim$ is **reflexive** if $x \succsim x$ for every $x \in X$. (there's always a weak preference between a choice and itself).
- $\succsim$ is **monotonic** if $x \geq y$ implies $x \succsim y$ ¹
- $\succsim$ is **strictly monotonic** if $x \geq y$ and $x_j > y_j$ for some j implies $x \succ y$.
- $\succsim$ is **convex** if for all x, y , and z in X such that $x \succsim z$ and $y \succsim z$, then $tx + (1 - t)y \succsim z$ for all t in $[0, 1]$.

1.2 Utility Function

Definition: A utility function $u : X \rightarrow \mathbb{R}$ represents $\succsim$ if, for every pair of points x and y in X , $x \succsim y$ if and only if $u(x) \geq u(y)$. In other words, if x has higher utility than y , you prefer x to y .

- **Theorem (Existence):** If preference relation $\succsim$ is rational and continuous on set X , there exists a continuous utility function $u : X \rightarrow \mathbb{R}$ that represents it.
- **Non-uniqueness:** The utility function representing preferences is not unique. If $u(x)$ represents preferences $\succsim$ and f is a strictly increasing function, then $f(u(x))$ represents $\succsim$ as well. For most of the time we only care about the **ordinal** property of utility function.

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¹If $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ are n -dimensional vectors, $x \geq y$ means that $x_j \geq y_j$ for $j = 1, \dots, n$

2 Indifference Curves

Let's stick to the 2-good case (x and y are our goods). When this happens, indifference curves take the following form for some constant c :

$$\begin{aligned}u(x, y) &= c \\ \Leftrightarrow u(x, y) - c &= 0\end{aligned}$$

Take the latter expression, which is set equal to 0. We can treat y as our 1 unknown variable for this single equation and x as a parameter. (In other words, think of this expression as saying $h(y; x) = u(x, y) - c$). By IFT, if the above expression is differentiable and that $\frac{\partial u(x, y)}{\partial y} \neq 0$, then:

- we can write y as a function of x and our constant c
- we can determine $\frac{dy}{dx}$ - note that the negative of this quantity is called the **marginal rate of substitution** (MRS). Specifically, $\frac{dy}{dx} = -\frac{\partial u / \partial x}{\partial u / \partial y}$. You can think of the MRS as how much of y you would need in exchange for x in order to stay at the same utility. Visually, this would be the negative of the slope of an indifference curve.

3 Exercises

3.1 Preferences and Utility Functions

Suppose there are two goods in the world: apples (x) and oranges (y), and that our preferences can be represented by the following utility function: $u(x, y) = 2x + y$

1. If a consumer does not have any apples, how many oranges would he need to be indifferent to a consumption bundle containing 2 apples and 1 orange?
2. Draw the indifference curves for when the utility equals 5 units and when the utility equals 7 units. Do these indifference curves cross? Can they ever cross? Why or why not?
3. Prove that the preferences represented by this utility function are transitive. That is, show that if $(x_1, y_1) \succsim (x_2, y_2)$ and $(x_2, y_2) \succsim (x_3, y_3)$, then $(x_1, y_1) \succsim (x_3, y_3)$.

3.2 Utility function

Consider the following utility function, with $\alpha > 0$ and $\beta > 0$:

$$u(x, y) = \alpha \log(x) + \beta \log(y)$$

Recall the definition of **monotonic preferences** (basically, *more is not worse*).

1. Are the preferences represented by the above utility function monotonic? Explain.
2. What is the sign of the slopes of our indifference curves? Are they upward sloping or downward sloping?
3. Let's consider another utility function, the utility function: $\tilde{u}(x, y) = x^\alpha y^\beta$. Prove that the two utility functions u and $\tilde{u}$ represent the same preferences; that is, prove that

$$u(x_1, y_1) \geq u(x_2, y_2) \Leftrightarrow \tilde{u}(x_1, y_1) \geq \tilde{u}(x_2, y_2)$$

4. Suppose currently market price of x is p , price of y is 1. Currently consumer is consuming at the point (x_0, y_0) . What would consumer do if $MRS(x_0, y_0) > p$? What would consumer do if $MRS(x_0, y_0) < p$?
5. Suppose now the utility function has changed to $u(x, y) = 2x + y$. How would your answer in Part (4) differ?

4 A Brief Review

4.1 Utility maximization

Formally, our optimization problem is as follows in the two-good case:

$$\max_{x_1, x_2} u(x_1, x_2) \text{ s.t. } \underbrace{p_1 x_1 + p_2 x_2 \leq M}_{\text{budget constraint}}, \quad x_1 \geq 0, \quad x_2 \geq 0$$

In other words, we want to choose the quantities of goods x_1 and x_2 that maximize our utility, subject to the following constraints:

1. The bundle does not cost more than our income M .
2. We consume non-negative amounts of each good.

Budget Constraint: Check whether the **marginal utility** with respect to x_i , or $\frac{\partial u}{\partial x_i}$ is *strictly* positive for all values of (x_1, x_2) . If so, then preferences are *strictly monotonic* in x_i (meaning *more is better*). In this case, a utility-maximizing individual will not leave any portion of M unspent, because she could always increase her utility by purchasing more of x_i .

After showing $\frac{\partial u}{\partial x_i} \geq 0$ for all x_i and for some j , we have $\frac{\partial u}{\partial x_j} > 0$, then our budget constraint will hold with equality, that is, $p_1 x_1 + p_2 x_2 = M$. In which case, you write the normal Lagrangian:

$$\mathcal{L}(\lambda, x_1, x_2; p_1, p_2, M) = u(x_1, x_2) - \lambda(p_1 x_1 + p_2 x_2 - M)$$

Positivity Constraints: We deal with these by trial-and-error. Ignore them when you write the Lagrangian, but once you have solved for x^* , we have to check whether the solution satisfies $x_i \geq 0$ for all i . If it doesn't, then x^* is NOT a solution to the maximization problem and you need to check other possibilities, namely corner solutions (when you spend all of M on one good and 0 on the other).

Solution:

- What's the graphic intuition of FOC?
- Can we have interior solution for convex utility functions? Why?

4.2 Indirect Utility Function

• **Definition:**

The indirect utility function takes prices and income as its arguments and gives you the maximum achievable utility. Here is the two-dimensional case (2 prices, 2 goods):

$$v(p, M) = u(x_1^*(p, M), x_2^*(p, M); p, M)$$

• **Properties:**

1. As we increase M the set of affordable bundles (the budget set) increases. Maximizing a function over a larger set will only increase the value at the optimum. Thus, $\partial v(\mathbf{p}, M) / \partial M \geq 0$.
2. As we increase p_i , the set of affordable bundles (the budget set) decreases. Maximizing a function over a smaller set will only decrease the value at the optimum. Thus, $\partial v(\mathbf{p}, M) / \partial p_i \leq 0$.
3. Note that we can use the Envelope Theorem to find how our (indirect) utility changes with prices and our income. For example:

$$\frac{\partial v(p, M)}{\partial p_1} = \frac{\partial \mathcal{L}(\lambda^*, x_1^*, x_2^*; p_1, p_2, M)}{\partial p_1} = \frac{\partial u(x_1^*, x_2^*)}{\partial p_1} - \lambda^* x_1^*$$

Our prices usually don't appear in the utility function, in which case, the above expression reduces to the following: $\frac{\partial v(p, M)}{\partial p_1} = -\lambda^* x_1^*$.

Likewise, $\frac{\partial v(p, M)}{\partial M} = \lambda^*$. Thus we can interpret λ as marginal utility of an extra dollar of consumption expenditure, or marginal utility of "income".

5 Exercises

5.1 Review of basic utility functions

Derive MRS for the following four utility functions, roughly draw the indifference curves, write down Marshallian demand functions of x and y .

- Cobb-Douglas: $u(x, y) = x^\alpha y^\beta$
- Stone-Geary: $u(x, y) = \alpha \log(x - c) + \beta \log y$
- Perfect substitute: $u(x, y) = \alpha x + \beta y$
- Perfect complement: $u(x, y) = \min(\alpha x, \beta y)$
- CES: $u(x, y) = (\alpha x^\rho + \beta y^\rho)^{\frac{1}{\rho}}$

5.2 Midterm 1, 2008

Yassine likes three goods: consumption goods x_1 , x_2 , and leisure l . He maximizes the utility function $u(x_1, x_2, l) = x_1^{\alpha_1} x_2^{\alpha_2} l^\gamma$ With $0 < \alpha_i < 1$ for $i = 1, 2$ and $0 < \gamma < 1$. The consumption good x_i has price p_i (for $i = 1, 2$), the hourly wage is w .

1. Consider that Yassine has H hours to work and, if he does not work, he takes leisure. For example, H could be 24 hours if the time period is a day. Hence, the hours worked h equals $H - l$. There are no sources of income other than the income from hours worked. Write down the budget constraint as a function of x_1 , x_2 , and l . [Hint: money spent on goods has to be smaller than or equal to money earned.]
2. Write down the maximization problem of the worker with respect to x_1 , x_2 , and l with *all* of the relevant constraints. Assume that the budget constraint is satisfied with equality. Why can we assume this? Provide as complete of an explanation as you can.
3. Write down the Lagrangian and derive the first order conditions with respect to x_1 , x_2 , l and λ .
4. Solve for x_1^* as a function of the prices p_1 , p_2 , w , the total number of hours H , and the parameters α_1 , α_2 , and γ . [Hint: combine the first two FOCs and then combine the first and third FOCs, and then finally, plug those into the budget constraint to solve for x_1^* . Similarly solve for x_2^* and l^* .]